



# Rights over software & EU data and cybersecurity legislation

Larisa Gabudeanu – ISM – 2024-2025

# Overview of topics

1. Definitions
2. Copyright over software and databases
3. Company and its employees/individual contractors
4. Company and external software development company
5. Off-the-shelf software
6. Customising off-the-shelf software
7. Embedding third party source code/libraries into your source code
8. Best practices in NDAs (non disclosure agreements)
9. PETs
10. EU data and cybersecurity legislation
11. Cyber crime – main criminal offences

# 1 of 11: Definitions

- Software
  - Any form of an application, operating system or any type of computer program
  - In any computer language
  - In the form of source code, object code; and
  - Any preparatory material for creating the software and any manuals
- Database
  - A collection of data or any other independent items
  - Protected through copyright or not
  - Placed in a systematic manner
  - Accessible through electronic or other means
- Creator of a database
  - Natural or legal person
  - That made a substantial qualitative or quantitative in obtaining, verifying or presenting the content of the database

# 1 of 11: EXERCISE - Definitions

- Which of the above is protected by software copyright?
  1. A script that you use to compare if two entries from two databases are identical
  2. The Google Analytics script references in websites
  3. Cookies
  4. Macro written by you in excel
  5. Third party library you include in Python code
  6. App in Play Store for Android
  7. Design architecture
  8. User manual
  9. Implementation document outlining the features of the software

Write on Slack for one use case: [no. of use case] – [yes/no] – [reason why]

# 1 of 11: EXERCISE - Definitions

- Which of the below is a database?
  1. IMDB
  2. SIEM database
  3. A MongoDB database for a fintech
  4. An Oracle database for an online retail store
  5. Antivirus signature database
  6. CVEs database (<https://cve.mitre.org/> )

Write on Slack for one use case: [no. of use case] – [yes/no] – [reason why]

# 2 of 11: Copyright over software and databases

- What is covered by protection under the law?
  - Source code and supporting documentation
  - Databases
  - Not the programming languages
  - Not the functionality of software
  - Not the graphical interface
  - Not the ideas and processes at the core/basis of the software
  - Not the principles and mathematical concepts

# 2 of 11: Copyright over software and databases

- What actions are NOT permitted on copyrighted software?
  - Bare copying
  - Having the same structures and the same steps, in the same order, for the same issue (if a substantial part of the source code was copied in this manner) – debated in case law
  - Without right, permanent or temporary reproduction (partially or fully)
  - Without right, translate and/or change a software
  - Distribution or rental of the software
- What actions are permitted on copyrighted software?
  - Back-ups needed for using the software
  - Analysing and testing a software in order to understand its principles
  - Interoperability
  - Error correction

# 3 of 11: Company and its employees/individual contractors

- **Default:**
  - Company holds rights of software created by its employees upon its instructions/as per their job description
  - Company does not hold rights over software created by individual contractors upon its instructions
- **Best practice:** Clause in agreement to clarify situation
- **Cross-border issues** – generally depends on legislation in the countries involved
- **Collaborative approach**
  - Default (co-authors who can decide jointly on rights over software)
  - Usually express clauses in agreements



# 4 of 11: Company and external software development company

- Specifics for software development agreement:
  - Instructions/business and technical **requirements** – add method to address in agile
  - **Cooperation platform** used – whose has access to it/back-ups
  - **Skills** required from external developers; subcontracting specifics
  - **Timeline and acceptance steps** – add method to address in agile
  - **Copyright** over software to be detailed – including not final forms
  - Maintenance and dependencies on provider (**exit strategy**)
  - Source code in **escrow**?
  - **Warranty** – especially over ownership of source code included in software
  - Integration with **legacy systems** – needed skills, approaches and liability, unknown integration aspects
  - **Updates** – legal or business requirements
  - **Testing** – use cases, who performs the tests and acceptance
  - Use of **production data** or synthetic data
  - **Non-compete**
  - **Confidentiality clause** – especially about business or confidential data the external developer has access to

## 5 of 11: Off-the-shelf software

- Standard right to use the software
- Can be customised – changes made by entities having right to amend the software
- For integration, depending on the type of licensing (e.g. open source or not)

# 7 of 11: Embedding third party source code/libraries into your source code

- 4 software freedoms of the Free Software Foundation
  - *Freedom to use/run the software (freedom 0).*
  - *Freedom to analyse the software (freedom 1).*
  - *Freedom to redistribute the software (freedom 2).*
  - *Freedom to improve/change the software (freedom 3)*
- Copyleft: Mechanism to ensure freedoms for all users
  - use the same open source license on composite/derivative work and on redistribution
  - provide access to initial source code and to composite/derivative work
- Not possible to have “proprietary” software

# 7 of 11: Embedding third party source code/libraries into your source code

- **Technical benefits**
  - Faster development
  - Auditability
  - Interoperability
  - Easy to customise
- **Technical risks**
  - Immaturity of some OSS
  - Stability of new versions of the OSS
- **Business benefits**
  - Low costs
  - Fast development of new business lines and new products
  - New skill developed by developers
- **Business risks**
  - Potential vulnerabilities due to lack of maturity of OSS
  - Lack of guarantee

## 7 of 11: Embedding third party source code/libraries into your source code

| License Type      | Characteristic                                                                                                                                  | Examples                              |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Permissive        | <ul style="list-style-type: none"><li>- Few restrictions for reuse and redistribution</li><li>- May restrict derivative/composed work</li></ul> | BSD, MIT, Apache Software, License v2 |
| “Weak” copyleft   | Copyleft on the original work and not on derivative/composed work                                                                               | LGPL, MPL, CPL                        |
| “Strong” copyleft | Copyleft on all redistributed work and on derivative/composed works                                                                             | GPL2, GPL3, EURL                      |

## 8 of 11: Best practices in NDAs (non disclosure agreements)

- **Define the confidential data** protected through the NDA (trade secrets – specific legislation at EU level)
- **Define exclusions** – data publicly available, data obtained through other sources, data developed independently, data already known by the co-contractor
- Detail the **disclosure right** for the confidential data: auditors, authorities, shareholders
- Detail the persons **to whom the NDA applies** – e.g. all employees of the parties and obligations of the parties to ensure confidentiality from their employees
- NDA to be signed **before access to data is given** – together mainly with the data processing agreement
- **Monitor implementation** of the NDA – e.g. remote connections to be monitored during maintenance

### \*TRADE SECRET

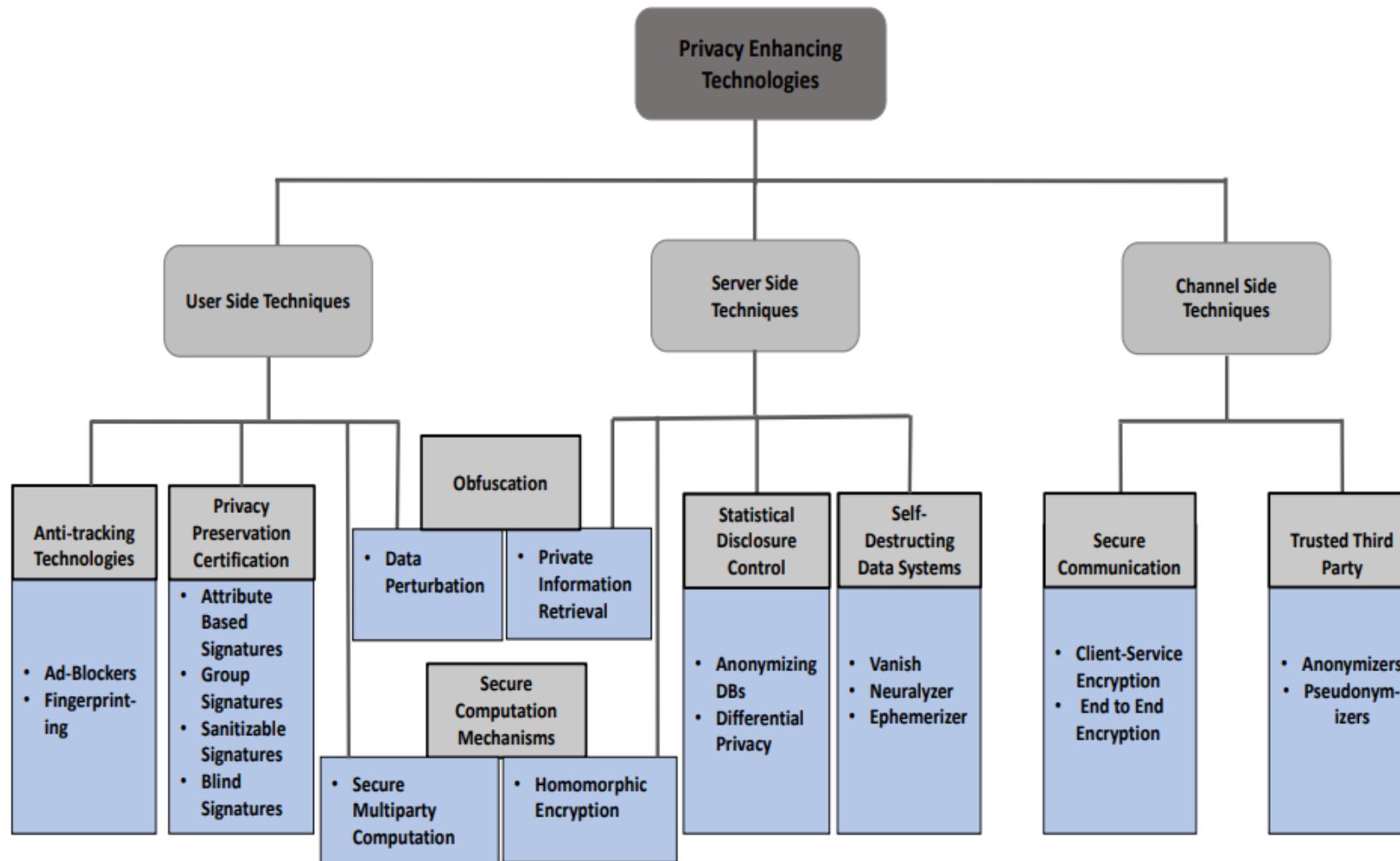
- Not generally known and not readily available to the persons who usually work in the area/field covered by the trade secret
- Has an economic value for the company invoking the trade secret
- The company having control over the trade secret has taken reasonable steps (under the given situation) to keep such information secret

## 8 of 11: EXERCISE - Best practices in NDAs

- What would you consider confidential during software development?
  1. UX/UI
  2. API format after it was made public for using by third parties
  3. Business requirements
  4. Source code
  5. Budget for the project
  6. Technologies used for development
  7. Technologies used for legacy systems with which the new software will interconnect

Write on Slack for one use case: [no. of use case] – [yes/no] – [reason why]

## 9 of 11 PETs - Purpose: protect personal data





# 9 of 11 PETs

- Purpose: protect personal data
- Examples – Top 10 PETs in 2023:
  - **Homomorphic Encryption** - encryption method enabling computational operations on encrypted data.
  - **Secure multi-party computation** - a subfield of homomorphic encryption. The difference is that users are able to compute values from multiple encrypted data sources.
  - **Differential privacy** - cryptographic algorithm includes a “statistical noise” layer to the dataset; it enables to describe patterns of groups within the dataset while maintaining the privacy of individuals.
  - **Zero-knowledge proofs** - uses a set of cryptographic algorithms that allow information to be validated without revealing data that proves it.
  - **Obfuscation**
  - **Pseudonymization** – examples: <https://ico.org.uk/media/1061/anonymisation-code.pdf>
  - **Data minimisation**
  - **Communication anonymizers** - replace online identity (IP address, email address) with disposal/one-time untraceable identity
  - **Synthetic data generation**
  - **Federated learning** - machine learning technique that trains an algorithm across multiple decentralized edge devices or servers holding local data samples, without exchanging them

10 of 11

## EU data

# and cybersecurity legislation



**Key:**  
Shaded =  
proposal or  
draft  
legislation

# 10 of 11 EU data and cybersecurity legislation

- **NIS Directive** – Romanian implementation – examples of provisions
  - ❖ Entities in scope for operators of essential services – e.g. from the banking sector, transportation, health sector, energy sector
  - ❖ Obligations include: implementation of security measures
- **DORA** (Digital Operational Resilience Act) – examples of provisions
  - ❖ Third party risk assessments
  - ❖ External red teaming exercise
  - ❖ ICT risk assessment
  - ❖ Resilience testing
- **Cloud** use in financial services legal provisions – EIOPA, EBA
- **Internet banking** local requirements for authorisation
- **Remote identification of individuals** – local requirements for authorisation

# 11 of 11 Cyber crime - main criminal offences

Art. 360 Cod Penal: Accesul ilegal la un sistem informatic

- (1) Accesul, fără drept, la un sistem informatic se pedepsește cu închisoare de la 3 luni la 3 ani sau cu amendă.
- (2) Fapta prevăzută în alin. (1), săvârșită în scopul obținerii de date informatice, se pedepsește cu închisoarea de la 6 luni la 5 ani.
- (3) Dacă fapta prevăzută în alin. (1) a fost săvârșită cu privire la un sistem informatic la care, prin intermediul unor proceduri, dispozitive sau programe specializate, accesul este restricționat sau interzis pentru anumite categorii de utilizatori, pedeapsa este închisoarea de la 2 la 7 ani.

Relevant aspects – **Illegal access to IT systems**:

- **Action sanctioned: Accessing**
- **Without right** – e.g. without a contract, without the consent of the owner of the IT system
- **IT system** (definition: orice dispozitiv sau ansamblu de dispozitive interconectate sau aflate în relație funcțională, dintre care unul sau mai multe asigură prelucrarea automată a datelor, cu ajutorul unui program informatic)
- **Purpose of accessing – relevant for increased sanctions**
- **Use of specialized tools – relevant for increased sanctions**
- **Attempts are sanctioned**

# 11 of 11 Cyber crime - main criminal offences

Art. 361: Interceptarea ilegală a unei transmisii de date informatice

(1) Interceptarea, fără drept, a unei transmisii de date informatice care nu este publică și care este destinată unui sistem informatic, provine dintr-un asemenea sistem sau se efectuează în cadrul unui sistem informatic se pedepsește cu închisoarea de la unu la 5 ani.

(2) Cu aceeași pedeapsă se sancționează și interceptarea, fără drept, a unei emisii electromagnetice provenite dintr-un sistem informatic, ce conține date informatice care nu sunt publice.

Relevant aspects – **Illegal interception of data transmission:**

- Action sanctioned: Interception
- Transmission of data that is not public
- Attempts are sanctioned

# 11 of 11 Cyber crime - main criminal offences

Art. 362: Alterarea integrității datelor informatice

Fapta de a modifica, șterge sau deteriora date informatice ori de a restricționa accesul la aceste date, fără drept, se pedepsește cu închisoarea de la unu la 5 ani.

Relevant aspects – **Altering the integrity of data:**

- Action sanctioned: modification, deletion or corruption/deterioration of data, restricting access to data
- Without right – e.g. without contract or consent
- Attempt are sanctioned

# 11 of 11 Cyber crime - main criminal offences

## Art. 363: Perturbarea funcționării sistemelor informatice

Fapta de a perturba grav, fără drept, funcționarea unui sistem informatic, prin introducerea, transmiterea, modificarea, ștergerea sau deteriorarea datelor informatice sau prin restricționarea accesului la date informatice, se pedepsește cu închisoarea de la 2 la 7 ani.

### Relevant aspects – **Perturbing the functioning of IT systems:**

- Action sanctioned: Perturbing the functioning of an IT system through input of data, transfer, change, deletion or deterioration of data
- Attempts are sanctioned

# 11 of 11 Cyber crime - main criminal offences

Art. 364: Transferul neautorizat de date informatice

Transferul neautorizat de date dintr-un sistem informatic sau dintr-un mijloc de stocare a datelor informatice se pedepsește cu închisoarea de la unu la 5 ani.

Relevant aspects – **Unauthorised transfer of data:**

- Action sanctioned: Unauthorised transfer
- Attempts are sanctioned



# Relevant legislation

- Law no. 8/1996 on copyright
- Emergency Ordinance no. 25/2018 on trade secrets
- Law no 11/1991 on protection from unfair competition
- Romanian Criminal Code
- Law no 362/2018 – implementing the NIS1.0 directive
- Order no 1323/2020 – security measures NIS1.0 directive



# Questions?

Email: [larisagabudeanu@gmail.com](mailto:larisagabudeanu@gmail.com)